

DEEP DRAWING OF **TITANIUM ALLOY**

SUBJECT – METAL FORMING AND MACHINING

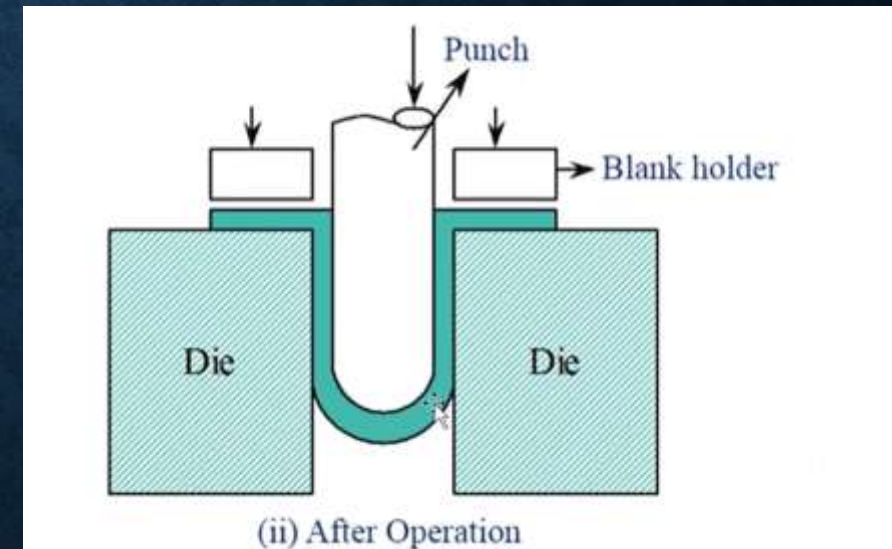
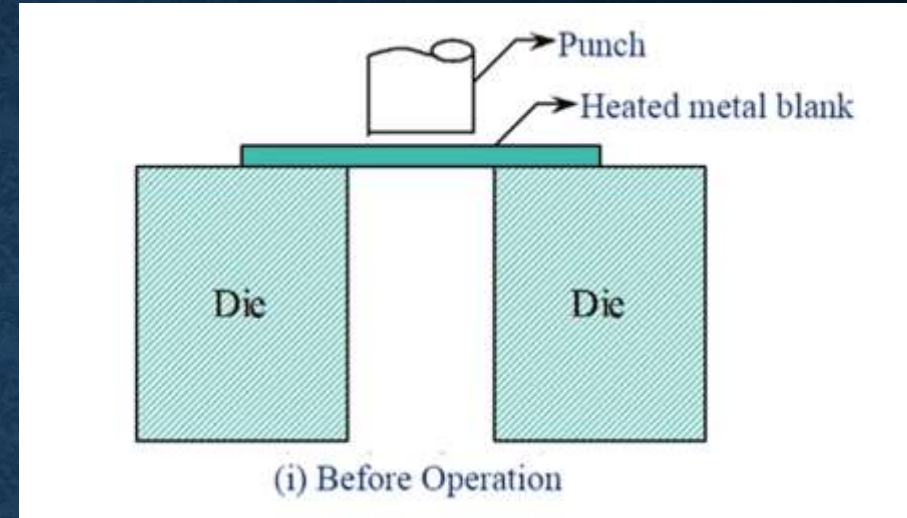
FACULTY NAME – DR. CHINMAYA PRASAD MOHANTY

STUDENT NAME – MD MANSOOR ALAM (24BME0562)

MUDDARGIRI VENKATA SIVA RAM (24BME0182)

INTRODUCTION-

- Deep drawing is a sheet metal forming process in which a flat sheet (called a blank) is mechanically pulled into a die cavity using a punch to create a hollow, cup-like or box-shaped component.
- It is widely used in manufacturing items like automobile panels, beverage cans, kitchen sinks, and aerospace components.
- The term “deep” is used when the depth of the drawn part is greater than its diameter.



Why did we choose Titanium?

- High Strength-to-Weight Ratio
As strong as steel, ~40% lighter, ideal for aerospace & automotive
- Excellent Corrosion Resistance
Forms protective oxide layer, works in marine & chemical environments
- High Temperature Capability
Retains strength at elevated temperatures, engine & aerospace parts
- Biocompatibility
Safe for medical implants & surgical tools
- Good Fatigue Resistance
Long life under cyclic loading

Challenge: Poor formability, high spring back, anisotropy, needs research

Applications of Deep Drawing Titanium –

- Aerospace - Fuel tanks, casings, structural shells (weight reduction)
- Automotive - Exhausts, lightweight parts (EV efficiency)
- Medical - Implants, surgical trays, prosthetics
- Chemical & Marine - Pressure vessels, heat exchangers
- Energy Sector - Nuclear & hydrogen system components

LITERATURE REVIEW

S.No	Author / Year	Journal	Paper title	Effect studied	Key findings
1	Zhao Zhao et al., 2024	Journal of Manufacturing Processes	Effect of forming temperature and fiber orientation on the warm deep drawing formability of TA2/Cf/PEEK laminates	TemperatureFiber orientation	Deep drawing slightly above T _g (200 °C) gives best forming quality. (0/45/90/−45°) pseudo-isotropic layup minimises boundary shrinkage anisotropy and delamination risk compared to unidirectional or orthogonal layups.
2	Kotkunde et al., 2014	Materials and Design	Experimental and numerical investigation of anisotropic yield criteria for warm deep drawing of Ti–6Al–4V alloy	Anisotropic yield criteria	Cazacu–Barlat criterion best captures HCP tension–compression asymmetry for Ti–6Al–4V at 400 °C, giving lowest average thickness-distribution error (1.6 %) and most accurate earing profile among five criteria tested.
3	Kotkunde et al., 2014	Materials and Design	Failure and formability studies in warm deep drawing of Ti–6Al–4V alloy	TemperatureFailure mode	LDR remains 1.8 from 150–400 °C, well below steel/Al. Two fracture modes identified: neck fracture (low temp) and upper-wall shear chipping (400 °C, large blanks). ANN–GA optimisation gives best uniform thickness at 219 °C, 54 mm blank.
4	Yu et al., 2025	Journal of Materials Research and Technology	Forming characteristics and precision control method of Ti60 titanium alloy ultra-thin hyperbolic shell in electro-assisted drawing	Electro-assisted formingSpringback	Continuous pulse current (AE+XD, 12 A/mm ² , 0.833 mm/s) keeps deformation temperature within 815–906 °C, reducing springback by ~74 % vs. no-current condition, and achieves dual-phase microstructure with only 3.6 % max thinning.
5	Harada et al., 2014	Procedia Engineering	Improvement of formability for multistage deep drawing of Ti-15V-3Cr-3Sn-3Al alloy sheet	Multistage drawingSeizure prevention	Oxide-coating via heat treatment (750 °C, 5 min) acts as a barrier against seizure in beta-Ti multistage drawing. Intermediate re-annealing at the 3rd stage extends formability to 6 stages; thickness reduction at corners stays ≤ 12 %.

6	Zhang et al., 2012	Transactions of Nonferrous Metals Society of China	Microstructure and texture of commercially pure titanium in cold deep drawing	Microstructure	Texture evolution	Plastic strain accommodated by both dislocation slip and deformation twinning across the drawn part. Twinning weakens recrystallisation texture (especially in drawing region), while over-drawing promotes slip dominance and texture strengthening.
7	Singh et al., 2018	Journal of Manufacturing Processes	Prediction of earing defect and deep drawing behavior of commercially pure titanium sheets using CPB06 anisotropy yield theory	Earing defect	Blank shape optimisation	CP-Ti LDR = 2.143 with 13.7 % earing height at diagonal direction. CPB06 model accurately predicts material flow and earing. Two-stage blank modification technique reduces earing by 83 % with only 6 % loss in minimum cup height; peak load drops 22 %.
8	Fan et al., 2025	Journal of Alloys and Compounds	Temperature-dependent crack behavior and microstructure evolution mechanisms in hot stamped Ti65 titanium alloy sheets	Temperature	Cracking behaviour	Formability improves 900→1000 °C (IE: 6.65→9.60 mm). At 900 °C cracks initiate in bulging region (insufficient thinning resistance); at 1000 °C fracture shifts to drawing–flange junction due to enhanced anisotropy and α/β microstructural mismatch. Prismatic slip dominates plasticity at both temperatures.
9	Yu et al., 2025 (Anisotropic behavior – electropulse)	Journal of Materials Research and Technology	Anisotropic behavior and mechanism in electropulse-assisted drawing (Ti60 – referenced in upload list)	Electropulse effect	Anisotropy	Electropulse application reduces flow stress and forming force significantly (up to ~73 % reduction in R-direction stress). AE+XD loading mode provides the most uniform temperature distribution and best balance of thinning uniformity and springback control across all current loading configurations tested.

OBJECTIVES-

The primary objective of deep drawing is to convert a flat sheet metal blank into a hollow, three-dimensional shape (such as cups, cylinders, or complex containers) without removing material, while maintaining structural integrity and dimensional accuracy.

Key Objectives:

- Achieve desired shape and depth with uniform thickness as much as possible.
- Minimize defects such as wrinkling, tearing, and excessive thinning.
- Ensure efficient material utilization (low wastage).
- Maintain surface finish and dimensional precision.
- Optimize forming parameters like punch force, blank holder pressure, and lubrication.

METHODOLOGY-

Working Principle of Deep Drawing

The process is based on plastic deformation of metal under compressive and tensile stresses.

Step-by-step working:

1. Blank Placement - A flat metal sheet (blank) is placed over the die opening.
2. Blank Holding - A blank holder (or pressure pad) applies force to prevent wrinkling during deformation.
3. Punch Action - A punch moves downward and pushes the blank into the die cavity.
4. Material Flow-The material flows radially inward and downward, undergoing:
 - Tensile stress in the wall region
 - Compressive stress in the flange region
5. Final Shape Formation-The sheet takes the shape of the die, forming a cup or hollow structure.

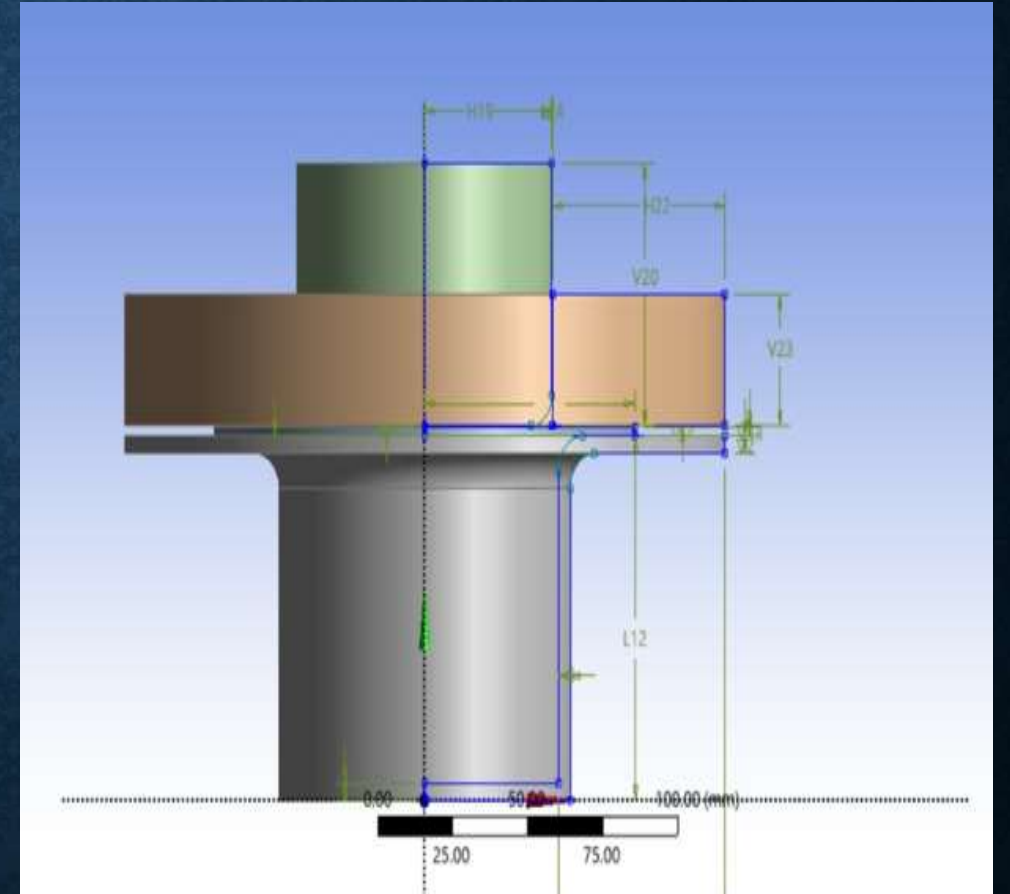
SIMULATION OBJECTIVES -

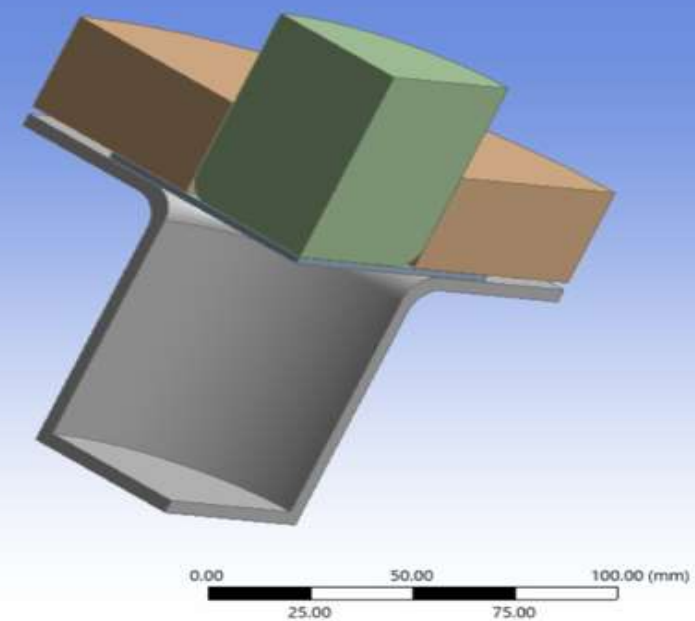
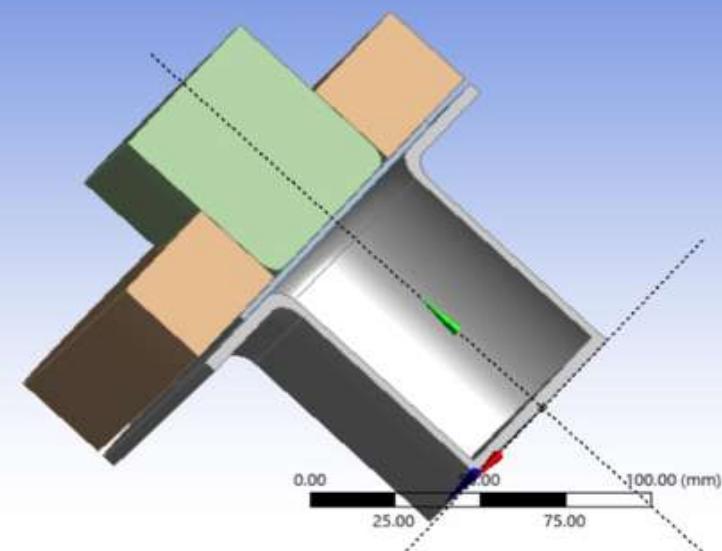
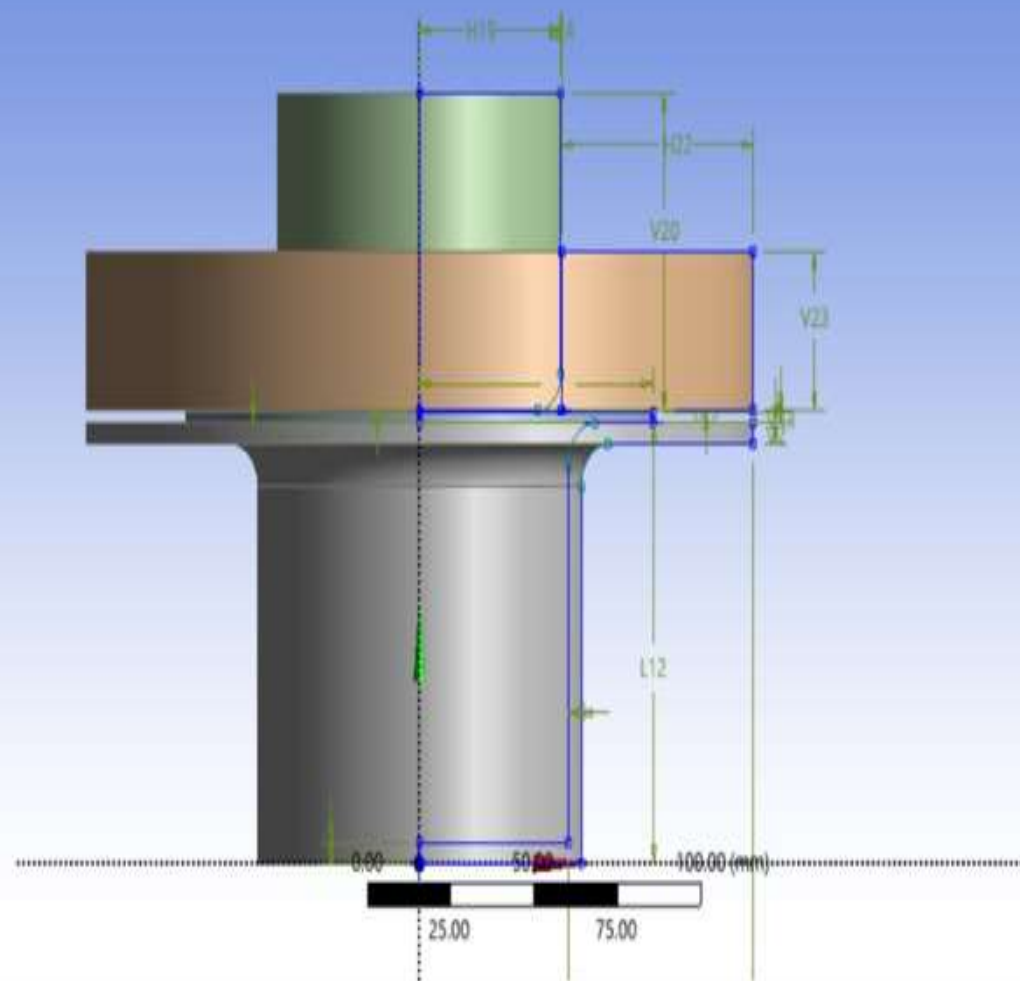
1. Geometry
2. Rigid and flexible bodies (only blank is flexible)
3. Sheet material (non linear titanium alloy)
4. Contact regions and Contact tools
 - Blank to punch
 - Blank to blank holder
 - Blank to die
5. Meshing
6. Load and Boundary Conditions (only for blank and punch in y dirn)
7. Analysis (Equivalent elastic strain, Equivalent stress, Equivalent plastic strain, Directional Deformation)

GEOMETRY-

Deep drawing geometry dimensions

Components	Dimensions in mm
Blank diameter	140
Blank thickness	2
Punch diameter	85
Punch height	60
Punch profile radius	7
Die diameter	89
Die height	83
Die width	200
Die thickness	4
Die profile radius	8
Blank holder height	30
Blank holder width	114.6





MESHING-

1. Discretization of the Geometry

The blank is divided into small finite elements (mesh), allowing the simulation to solve complex deformation using the **Finite Element Method (FEM)**.

2. Accurate Stress & Strain Prediction

Meshing helps calculate:

- Stress distribution
- Strain and thinning
- Areas prone to failure

3. Capturing Material Flow

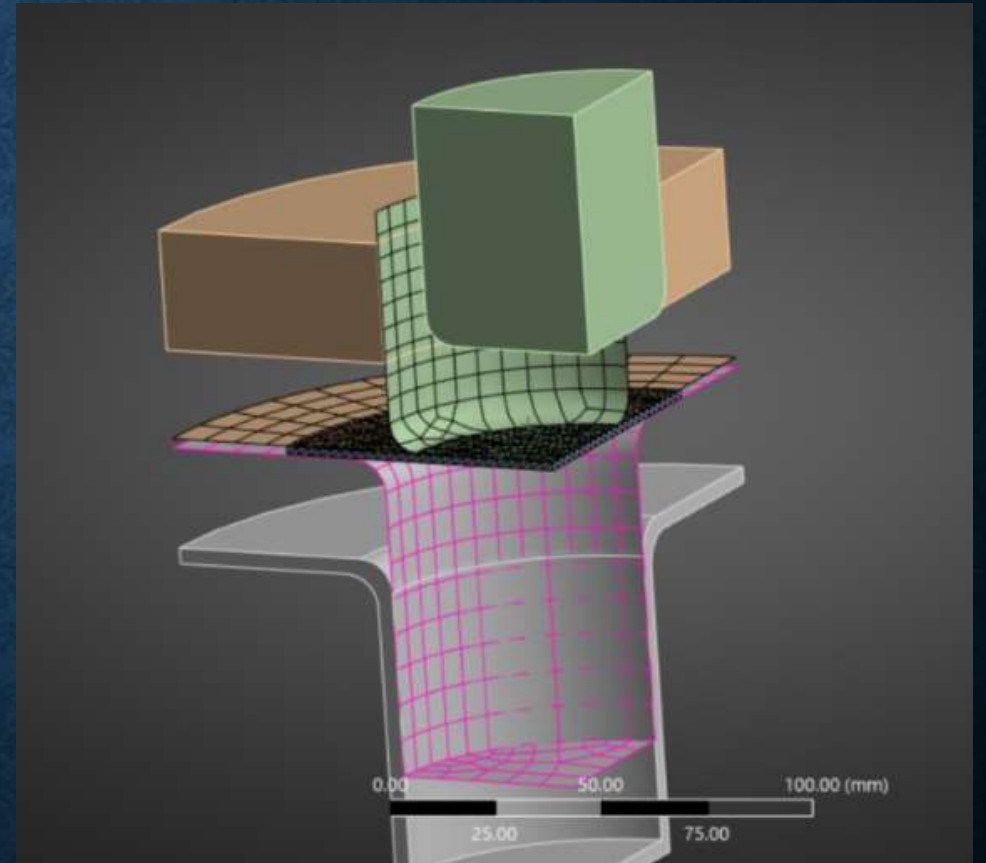
It shows how material flows from the flange into the die cavity—very important in deep drawing analysis.

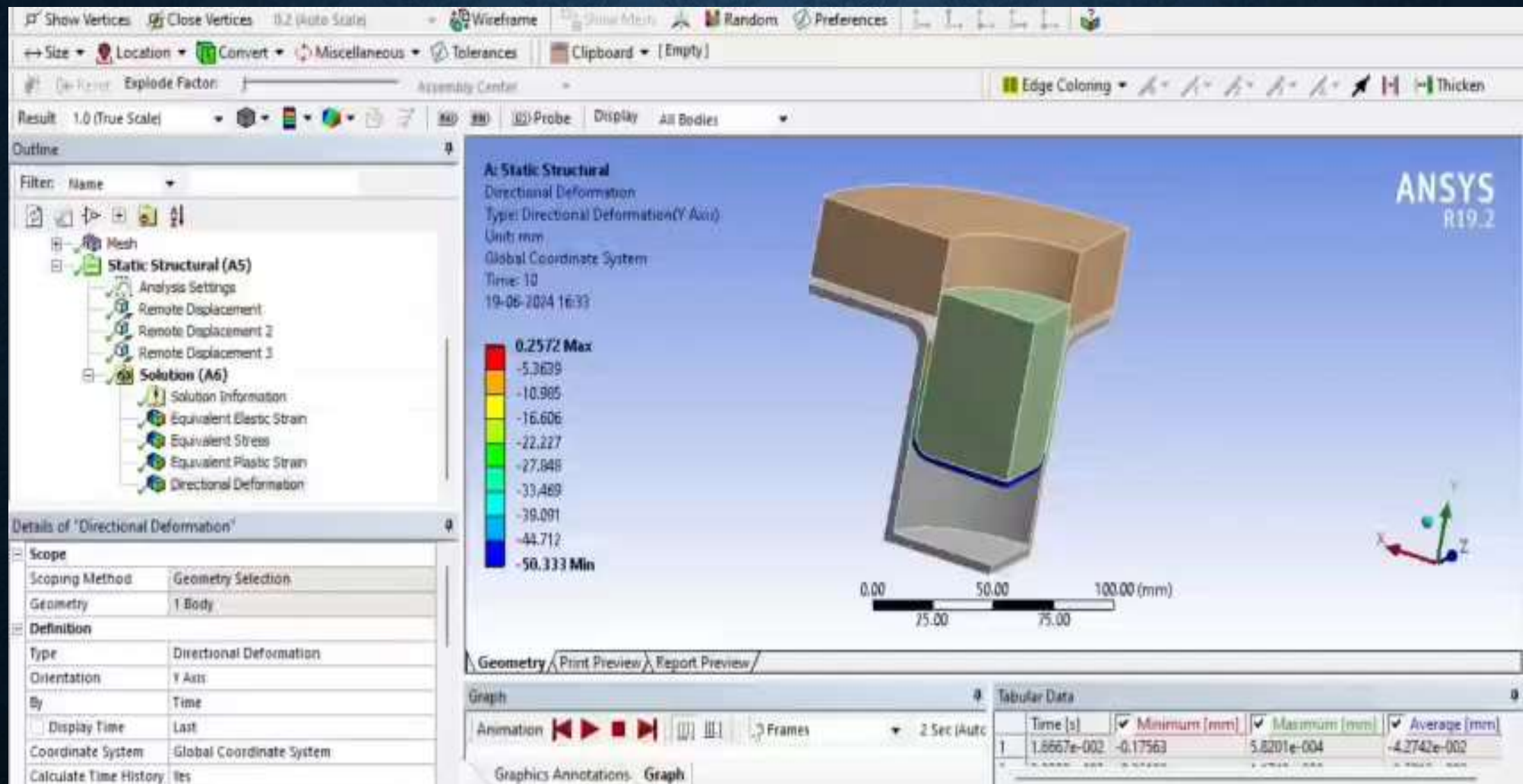
4. Defect Prediction

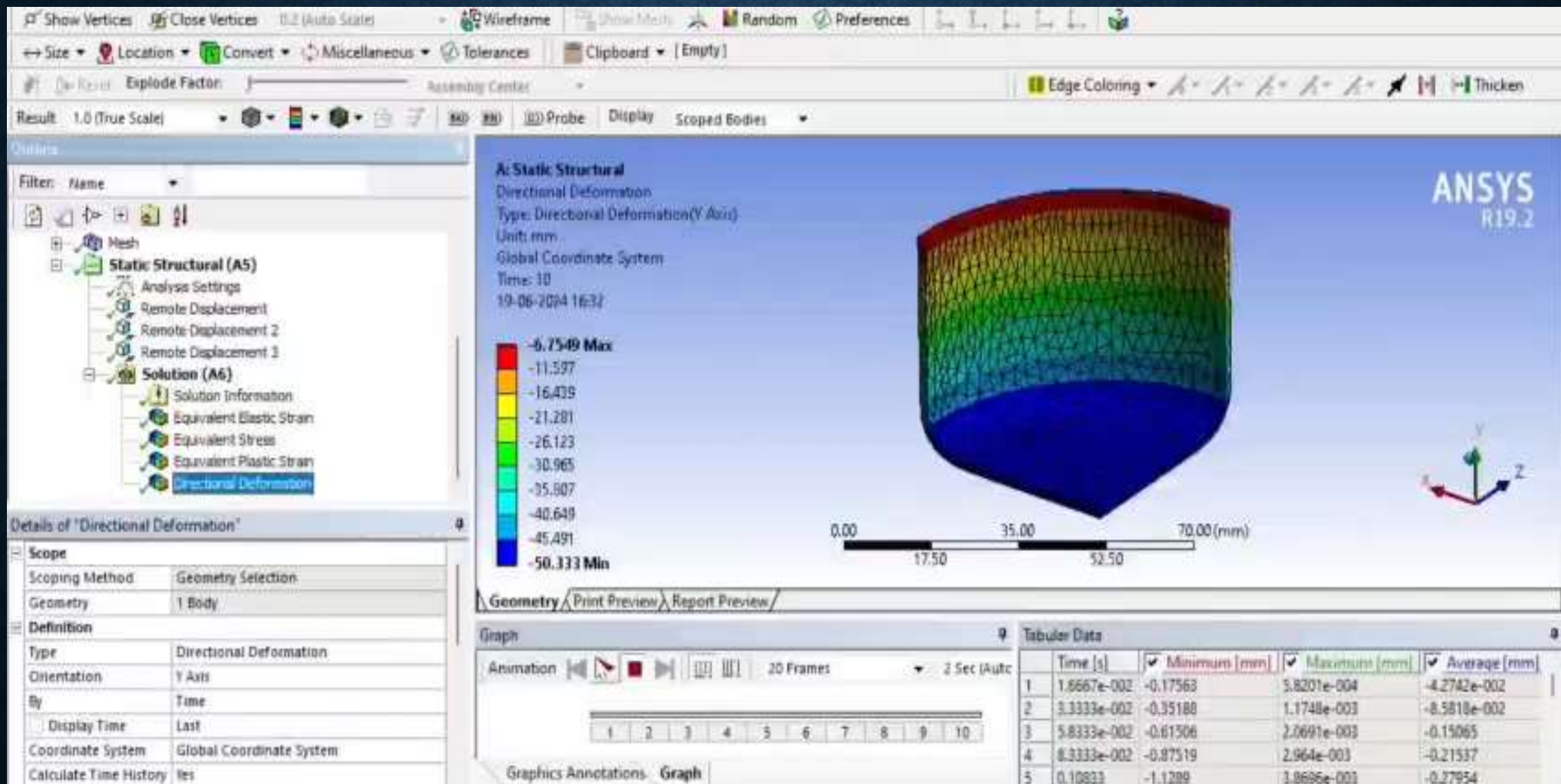
5. Improved Simulation Accuracy

- Fine mesh → higher accuracy (but more computation time)
- Coarse mesh → faster but less precise

6. Contact Interaction Handling







A: Static Structural

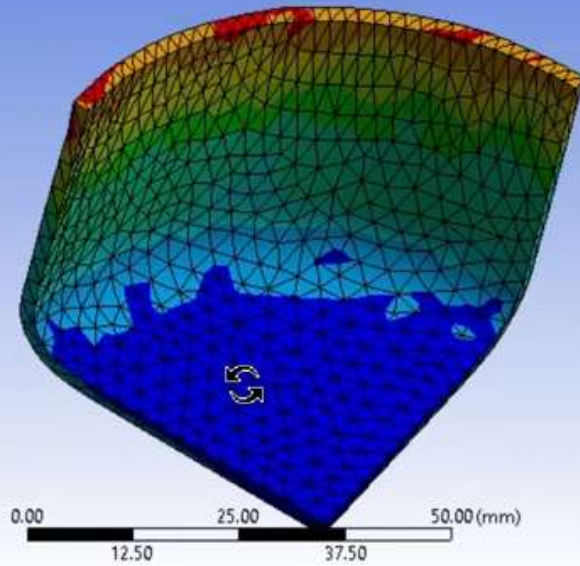
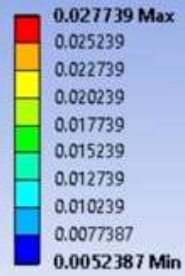
Equivalent Elastic Strain

Type: Equivalent Elastic Strain

Unit: mm/mm

Time: 10

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A: Static Structural

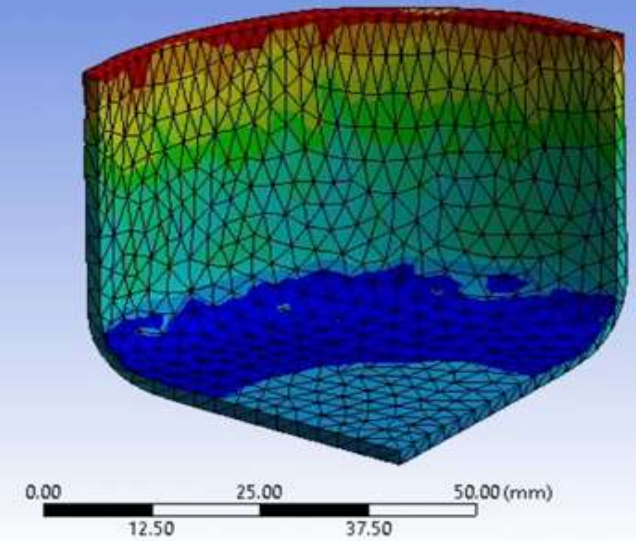
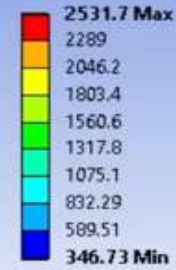
Equivalent Stress

Type: Equivalent (von-Mises) Stress

Unit: MPa

Time: 10

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A: Static Structural

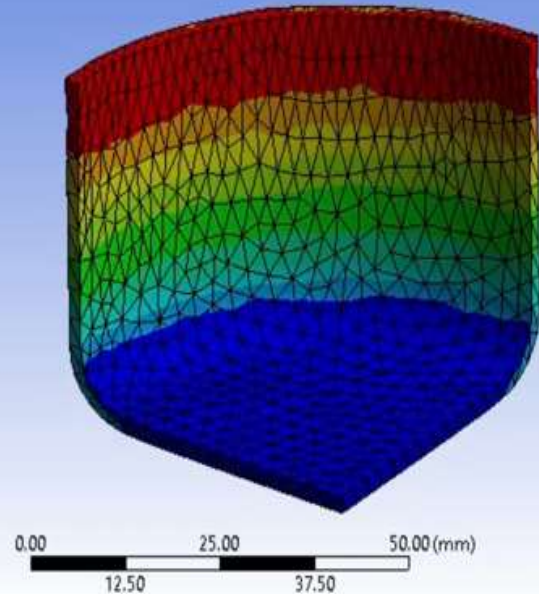
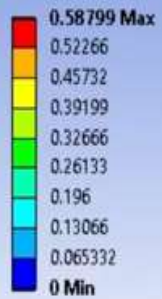
Equivalent Plastic Strain

Type: Equivalent Plastic Strain

Unit: mm/mm

Time: 10

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A: Static Structural

Directional Deformation

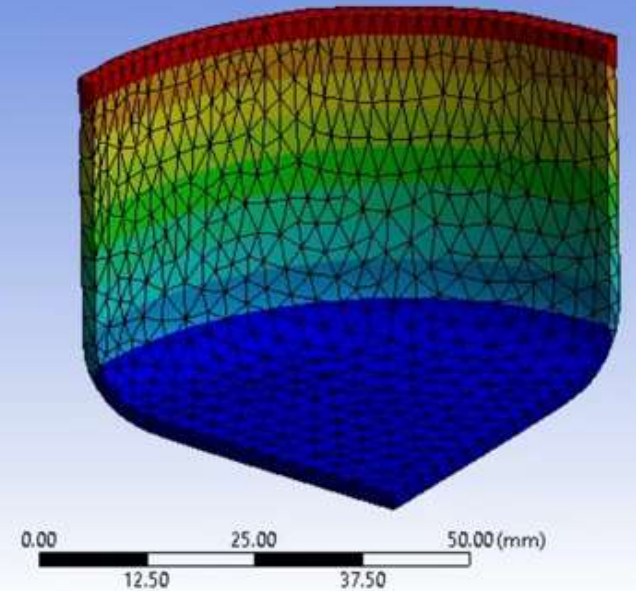
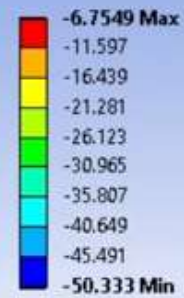
Type: Directional Deformation(Y Axis)

Unit: mm

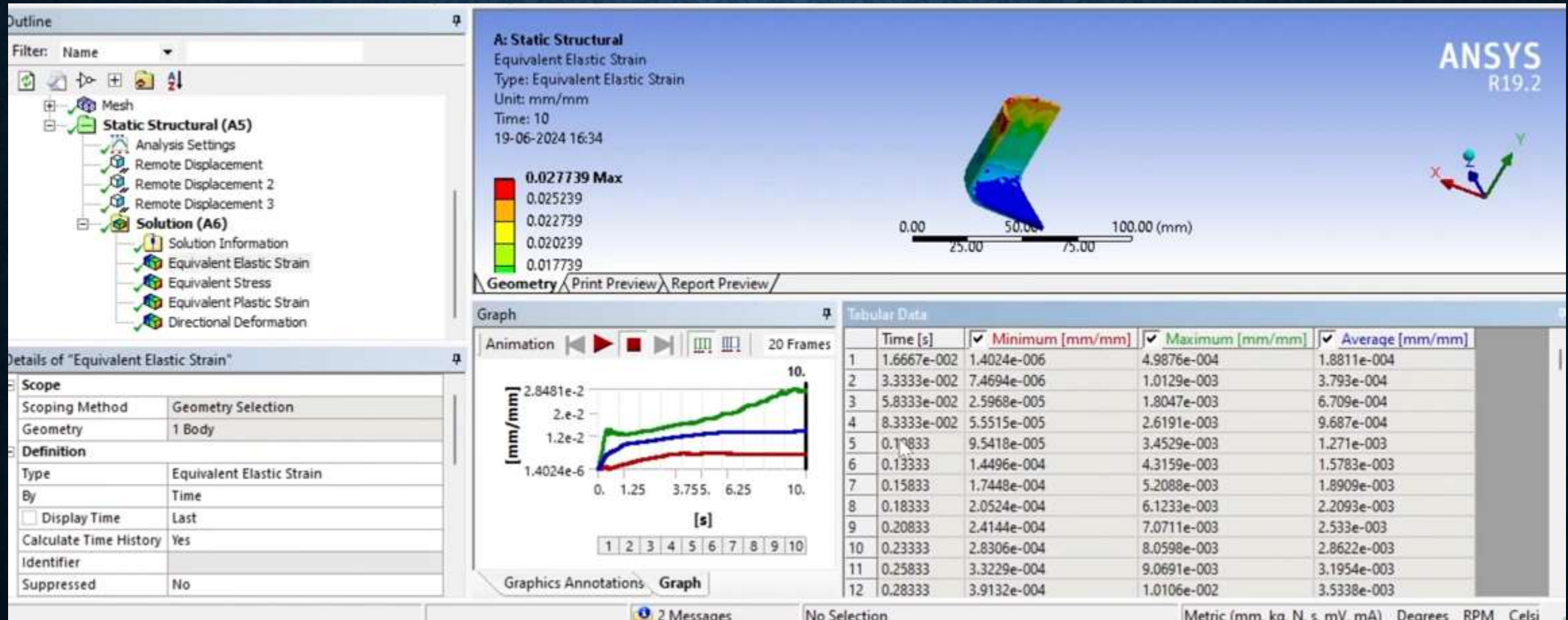
Global Coordinate System

Time: 10

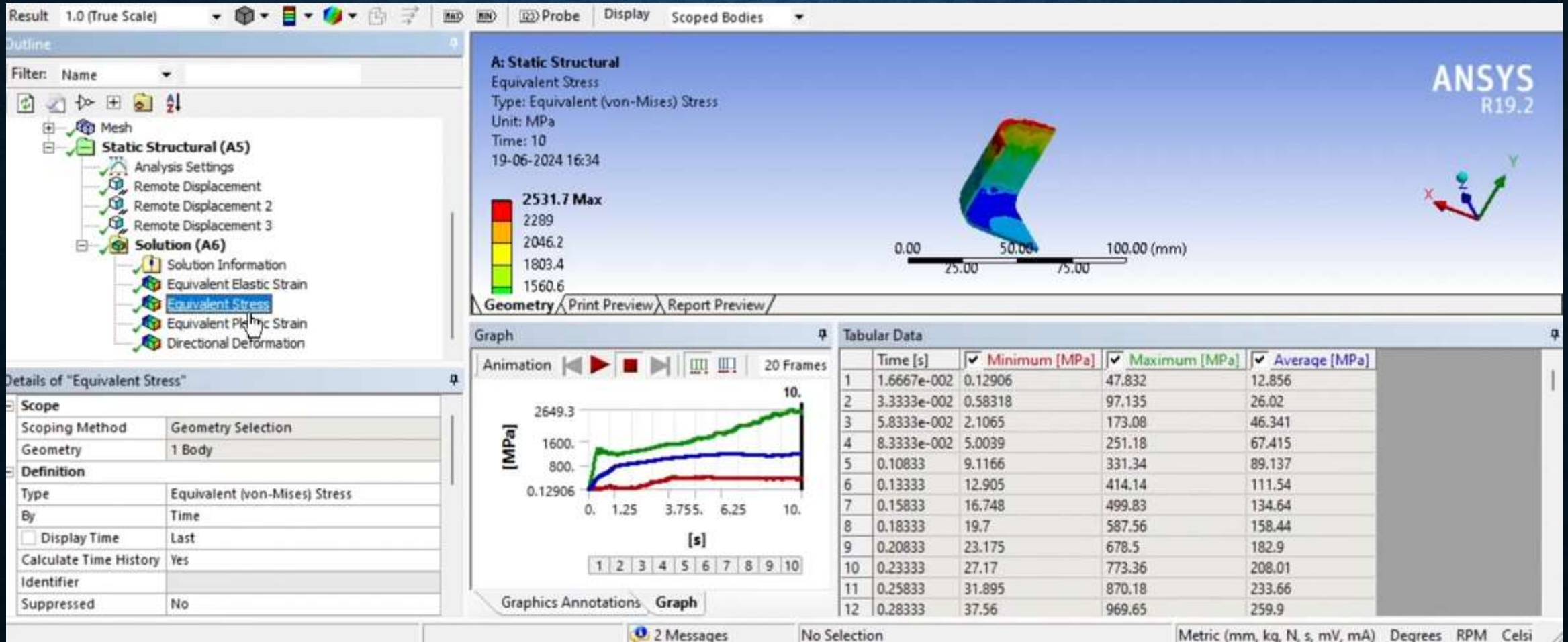
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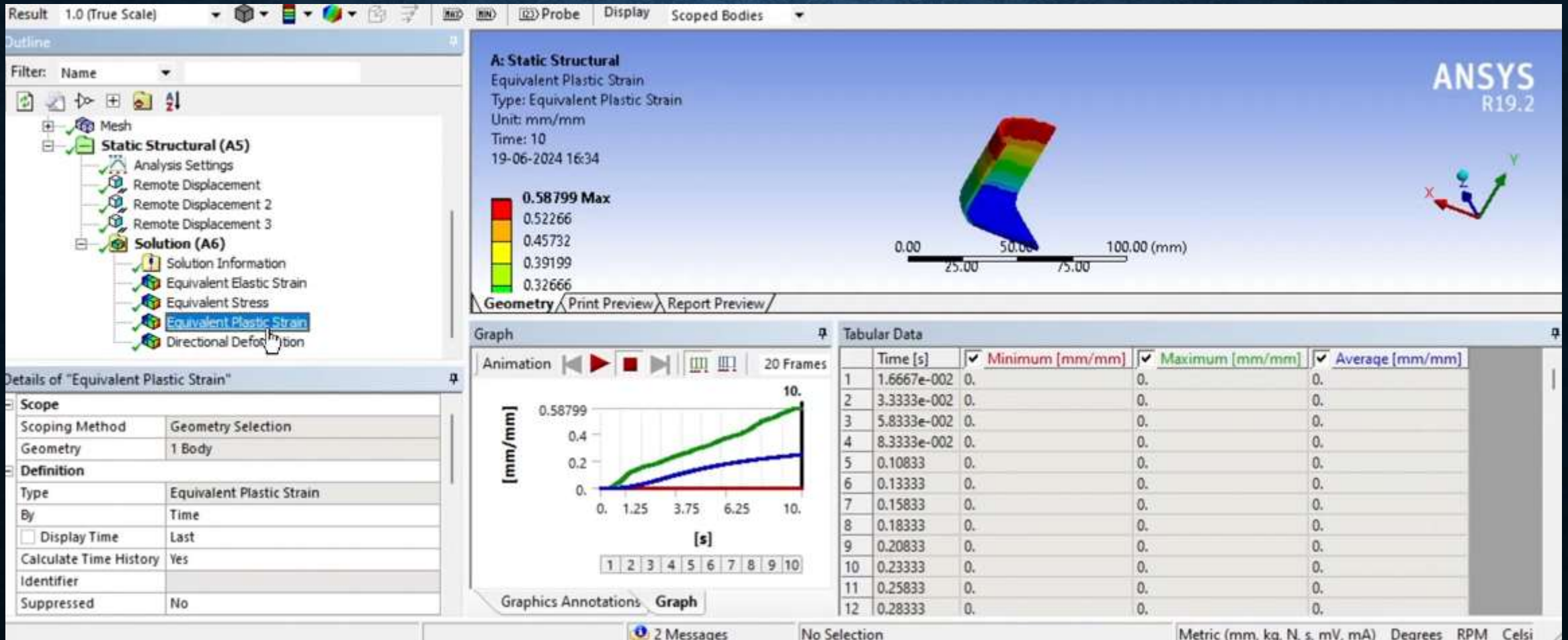
EQUIVALENT ELASTIC STRAIN -



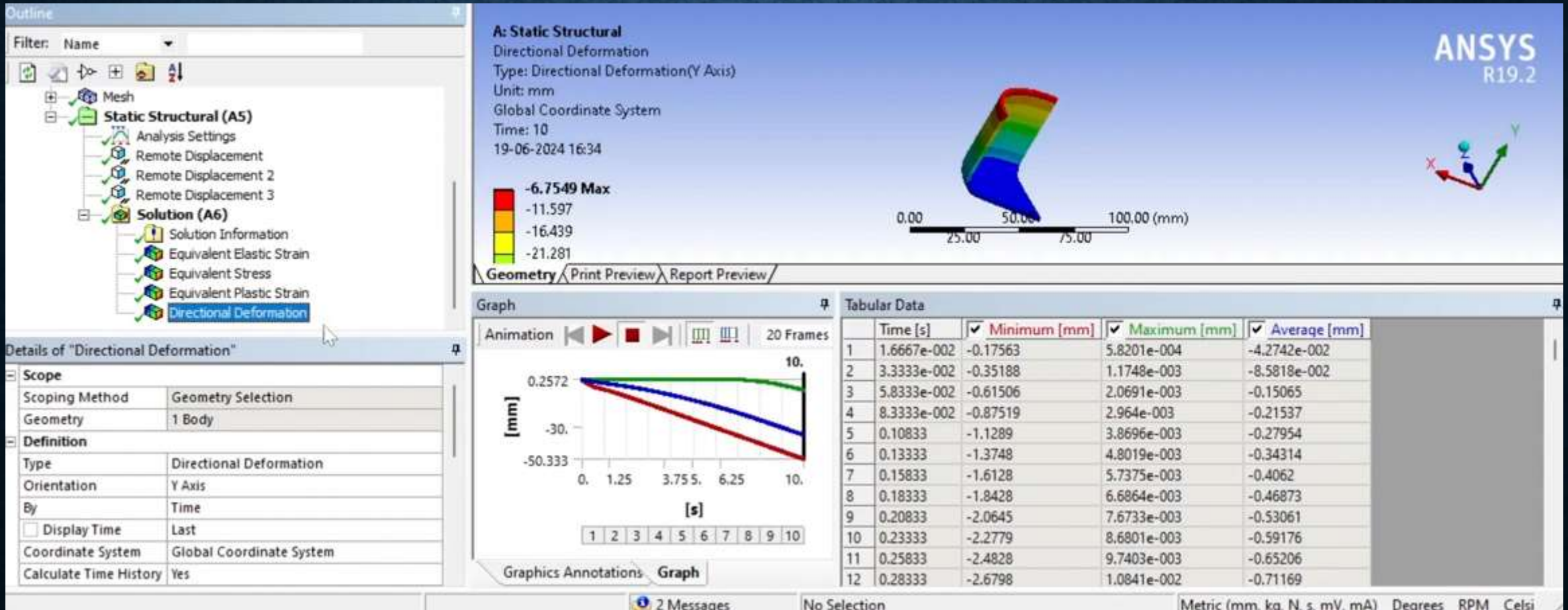
EQUIVALENT STRESS-



EQUIVALENT PLASTIC STRAIN



EQUIVALENT ELASTIC STRAIN



CONCLUSION-

- The study of deep drawing of titanium alloy demonstrates that complex hollow components can be successfully formed through controlled plastic deformation using appropriate tooling and process parameters.
- The simulation carried out in ANSYS Workbench provided valuable insights into material behavior, including stress distribution, strain variation, and deformation characteristics.
- The meshing of the blank played a crucial role in accurately predicting material flow, thickness variation, and potential defects such as wrinkling and tearing.
- The results from equivalent stress, elastic strain, and plastic strain analysis highlight that proper control of blank holder force, punch movement, and contact conditions is essential for achieving defect-free components.
- Overall, the study confirms that numerical simulation is an effective tool for optimizing deep drawing processes, reducing trial-and-error in manufacturing, and improving product quality, especially for advanced materials like titanium alloys

REFERENCES-

- <https://youtu.be/sOITyXXJB3Y?si=y4hX4OCGt7zDGc6b>
- <https://youtu.be/F2VTmUfk0AM?si=EoFlEzlDZ1QGo8Zh>
- <https://youtu.be/9OWzgJHh4Go?si=JNnXKlayyCDILJza>
- <https://youtu.be/UIl7rXMieFE?si=XC5DC60HEmZTKrLC>